CPR Chatbot Evaluation Report – Cycle 2

Introduction

This report presents the second evaluation cycle of the lifesaver CPR chatbot. The purpose of this cycle is to assess how the chatbot's performance changed after the developer implemented updates based on the findings from Cycle 1. Because the same questions and evaluation methods were used in both cycles, the results allow for a clear, fair comparison of progress over time.

The focus of this evaluation is on three core areas:

1. Readability — how easy the chatbot's responses are to understand during an emergency.
2. Content accuracy — whether the answers are correct, relevant, and aligned with CPR guidelines.
3. Thematic patterns — recurring types of errors that reveal deeper issues in the system's understanding and response generation.

Together, these measures provide a comprehensive view of how the chatbot improved, where it still struggles, and what changes are needed moving into Cycle 3.

Methodology

To ensure consistency with Cycle 1, the same 40 CPR-related questions were used to collect a new set of chatbot responses. The evaluation followed the same four-part methodology:

1. Readability Analysis

Each response was scored using the Flesch-Kincaid Grade Level. These scores were then rounded and grouped into readability categories (Low, Medium, High) to assess how accessible the language is for users in high stress situations.

2. Content Accuracy Assessment

Every response was manually labeled as Correct, Incomplete, or Irrelevant. This allowed for a direct comparison of accuracy between cycles and helped identify whether the chatbot's updates improved its ability to answer CPR-specific questions.

3. Thematic Analysis

Responses were categorized into recurring error themes, such as misunderstanding, irrelevant medical advice, template repetition, and misuse of child CPR instructions.

This step highlights deeper structural issues that may not be visible through accuracy scores alone.

4. Comparative Analysis

Cycle 2 results were compared directly with Cycle 1 to determine which issues improved, which persisted, and whether any new problems emerged after the update.

Results

Comparative Analysis of Cycle 1 and Cycle 2

This section presents a combined view of both cycles, highlighting how the chatbot's performance changed after the developer's improvements.

1. Readability Comparison

Readability improved modestly from Cycle 1 to Cycle 2.

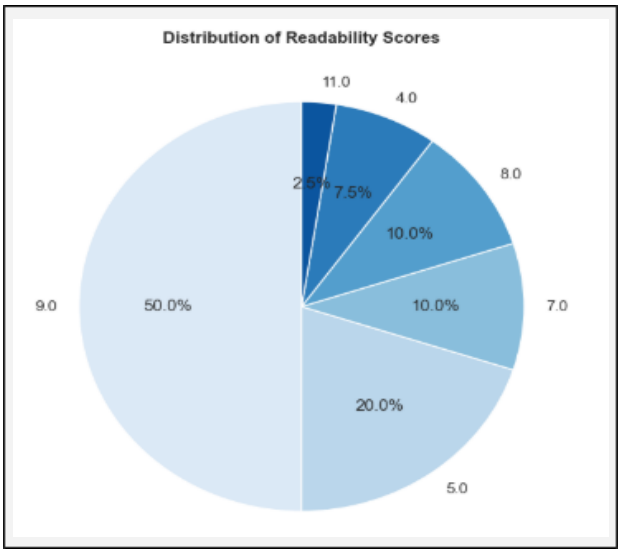
In Cycle 1, most responses were difficult to read:

- 67.5% fell into the Low readability category
- 32.5% were Medium
- None were High readability

Cycle 2 shows a shift toward more accessible language:

- 60% of responses were Medium readability
- 37.5% were Low
- 2.5% reached High readability for the first time

As shown in Figure 1, readability improved between cycles. .



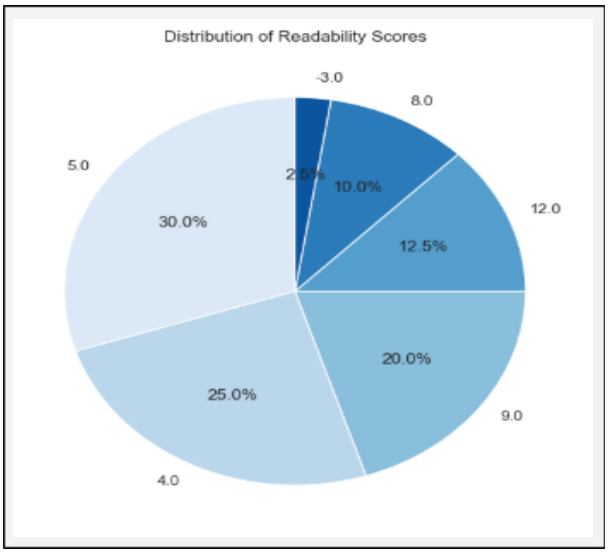
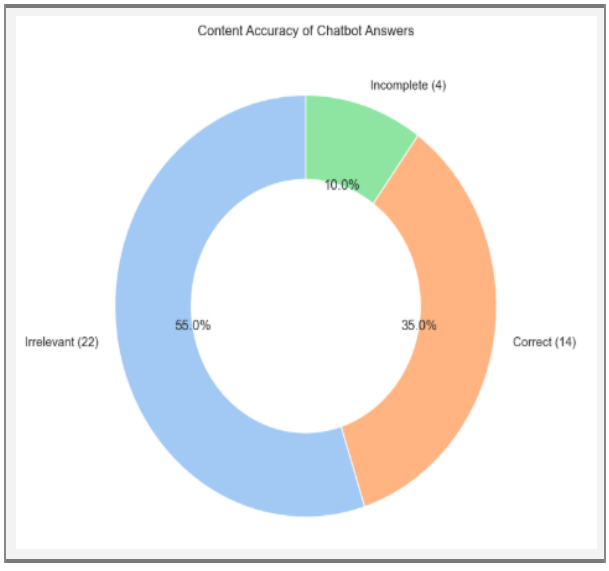


Figure 1. Readability Distribution – Cycle 1

Figure 2. Readability Distribution – Cycle 2

2. Content Accuracy Comparison

Content accuracy shows clearer improvement than readability as clearly shown in Figure 3 and 4.



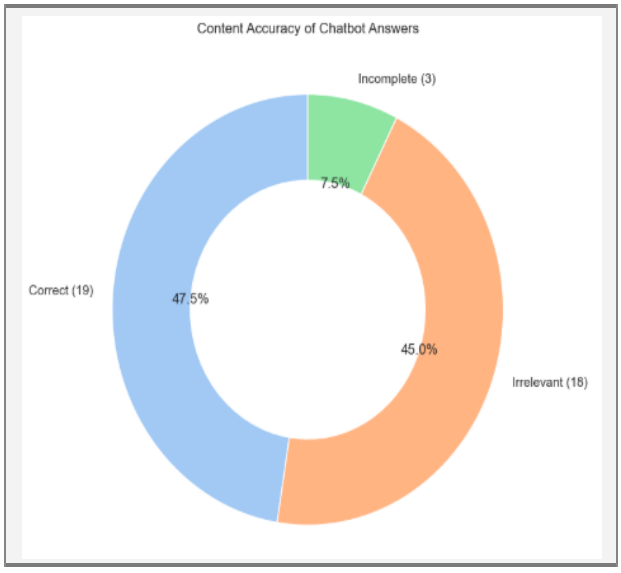


Figure 3.Content Accuracy – Cycle 1

Figure 4. Content Accuracy – Cycle 2

Cycle 1 was dominated by irrelevant answers, including hypothermia advice and conversational reassurance.

Cycle 2 demonstrates a meaningful increase in correct answers and a reduction in irrelevant ones. However, nearly half of the responses still fail to address the CPR question. Irrelevant content persists, now appearing as burn-treatment instructions instead of hypothermia, suggesting that the underlying misclassification issue remains unresolved.

3. Thematic Error Comparison

The thematic patterns across both cycles reveal where the chatbot improved and where it continues to struggle.

The figures 5 and 6 below illustrates the noticeable shift from Cycle 1 to Cycle 2, making the differences easy to see at a glance.

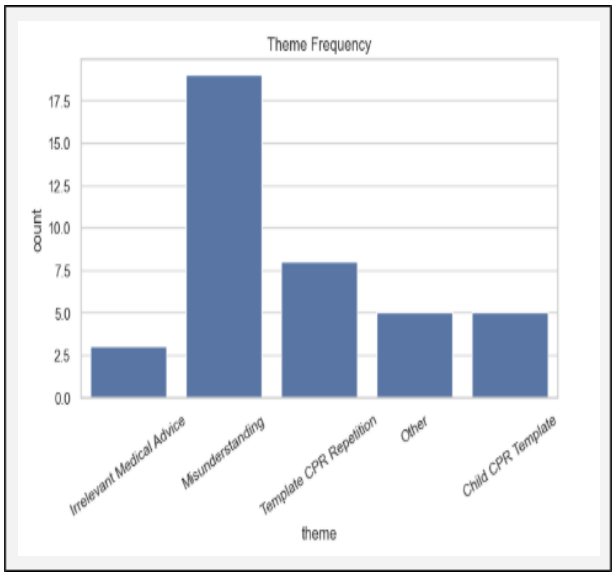


Figure 5.Theme Frequency – Cycle 1

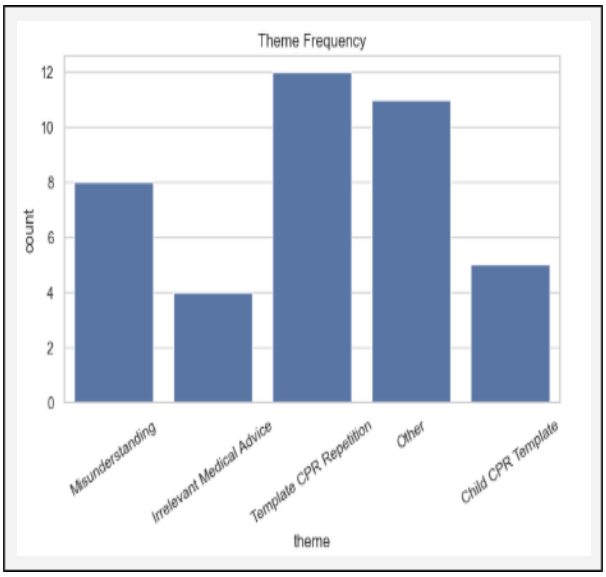


Figure 6. Theme Frequency – Cycle 2

- ☐ The both figures compares the frequency of error themes across Cycle 1 and Cycle 2.
- ☐ Cycle 1 shows “Misunderstanding” as the dominant issue.
- ☐ Cycle 2 shifts toward “Template CPR Repetition” as the most frequent error.
- ☐ Irrelevant medical advice appears in both cycles but changes in content (hypothermia → burn treatment).
- ☐ Child CPR template misuse and “Other” issues appear in both cycles at lower levels.

Overall, Cycle 2 eliminated some Cycle 1 issues, but the core structural weaknesses—misclassification, template overuse, and fallback errors persist.

4. Overall Performance Change

Across both cycles, the chatbot shows partial but meaningful improvement.

It produces more correct answers, fewer irrelevant ones, and slightly more readable responses. It also avoids the off-topic or conversational errors seen in Cycle 1.

However, major limitations remain. Nearly half of the responses in Cycle 2 are still irrelevant or incomplete. Template repetition and misclassification continue to affect reliability, and readability is still not consistently suitable for emergency use.

In short, the updates improved surface-level behavior, but deeper issues in understanding CPR-specific questions and selecting appropriate templates remain unresolved.

NLU Training Data Review

The Cycle 2 NLU structure still shows several weaknesses that directly contribute to the chatbot's errors.

1. CPR-Specific Intents Still Missing

Cycle 2 continues to rely on broad intents such as *how_to_cpr* and *what_is_cpr*.

There are still no intents for:

- adult CPR
- hands-only CPR
- compression depth
- compression rate
- rescue breaths
- AED pad placement
- when to start/stop CPR

This forces CPR questions into fallback or unrelated medical intents.

2. Non-CPR Intents Still Dominate

The NLU file still contains many non-CPR intents like burns….

These compete with CPR intents and cause misclassification.

3. Intent–Response Mapping Still Uneven

Many intents still lack dedicated templates, causing:

- fallback messages
- repeated CPR templates
- inconsistent answers

4. Adult vs Child CPR Still Confused

The domain still includes a child CPR template but no clearly labeled adult CPR template.

This leads to child CPR instructions appearing in adult scenarios.

5. Template Repetition Still High

Only one adult CPR template is used repeatedly, regardless of question context.

6. Fallback Still Overused

Fallback messages continue to appear due to missing CPR-specific intents.

Discussion

Cycle 2 shows meaningful improvement in accuracy and readability, but the core issues remain.

The chatbot still struggles to correctly interpret CPR-specific questions, often routing them into unrelated first-aid categories. Template repetition has increased, and irrelevant medical advice persists.

These issues indicate that the updates made after Cycle 1 improved surface-level behavior but did not address deeper structural problems in the NLU pipeline and response-selection logic.

Recommendations

N.B. The CPR-specific intents and structural fixes listed here were also recommended in the Cycle 1 report but were not implemented. To achieve meaningful improvement in Cycle 3, it is essential that all recommendations are fully applied.

To improve the chatbot's CPR performance, the following updates are recommended:

1. Add CPR-Specific Intents

Create new intents such as:

- adult_cpr_steps
- hands_only_cpr
- compression_depth
- compression_rate
- rescue_breaths
- aed_pad_placement
- when_to_start_cpr
- when_to_stop_cpr

Each intent should include **10–20 training examples**.

2. Add Clear Response Templates

Templates needed:

- Adult CPR
- Child CPR
- Infant CPR
- Hands-only CPR
- AED usage
- Start/stop CPR
- Troubleshooting

Templates must use:

- simple language
- bullet points
- clear steps

3. Improve Intent–Response Mapping

- Add rules that map each CPR intent to the correct template.

- Add stories for each CPR scenario.

4. Improve Topic Filtering

- Prevent CPR questions from triggering unrelated medical intents.
- Add disambiguation questions (e.g., “Is the victim an adult or a child?”).

5. Add Entity Recognition

Add entities for:

- age group
- breathing status
- consciousness

6. Reduce Template Repetition

- Add more varied CPR templates.
- Improve response-selection logic.

7. Reduce Fallback Errors

- Improve NLU classification.
- Add more CPR-specific training data.

Conclusion

Cycle 2 demonstrates clear progress compared to Cycle 1, especially in accuracy and readability. However, major issues remain, including misclassification, template repetition, irrelevant advice, and unclear adult vs child CPR differentiation.

By restructuring the NLU file, adding CPR-specific intents, improving templates, and simplifying language, the chatbot can become significantly more reliable and user-friendly in Cycle 3.